

LLM 4-dars.

Embedding is a numerical representations of data it turns complex, high-dimensional data into low-dimensional vectors.

Why Embeddings?

- Semantic Understanding: they capture semantic relationships between data points.
- Computational Efficiency: low-dimensional vector hõlqani u-u xishtlashni tezlatadi va samaradorlikni oshiradi.
- Generalization: Noise ni ignore qilish va kerakli featurelarga e'tibor berish orqali modelga yaxshi o'rganish imkonini beradi.

Tokenization is a process of breaking down text into smaller units called tokens, such as words, sub words or characters.

Turkari Embedding:

1. Input $\approx$ ^{Token} RoPE

2. Output $\approx$ Contextual $\approx$ positionni aniqlaydi

Why **RoPE**? $\rightarrow$ Sanoat Standarti, Long Context,
Relative + Absolute, +

masofani chunadi + qo'shish sodda
+ base frequency parametri (θ) qanday
sozlanishi context-length tengaytirishga
hissir qiladi.

Attention $\Rightarrow$ feature representation.

Faydalanish: Input Embeddings (Token + RoPE) $\rightarrow$
Transformer layer (in Attention)

Ma'nosiz, bog'lanmagan vectorlarni mo'noli qilib
bog'lash va uzun tekstlarni erab qolish

$Q \rightarrow$ Query

$K \rightarrow$ Key

$V \rightarrow$ Value

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V$$

Attention Türleri:

- Architectural Types (Cashemi qantarp tejashini bilasli)

* MHA $\rightarrow$ Multi Head Attention

MAQ $\rightarrow$ Multi Query Attention

Good $\leftarrow$ GQA $\rightarrow$ Grouped-Query Attention

- Context Optimization

- Sparse

- Sliding Window